F005 v0.5.0 — Independent Linearized-Euler Sloshing Baseline

Development science contract and numerical qualification

SST-Workbench / F005 External Wave Benchmark

21 October 2026

Status: GEOMETRY_UNVERIFIED for physical use. The code, analytical tests and synthetic demo are executable. This is a linear potential-flow solution of the *linearized* Euler free-surface problem; it is neither full nonlinear Euler computational fluid dynamics nor source-native PKLSA.

1 Question and evidence boundary

Given the published longitudinal tank acceleration $a(t)$, can an ideal inviscid fluid model derived *ex ante* from the tank's measured geometry predict longitudinal free-surface elevation independently of trained ARX/oscillator parameters? The Zenodo release describes a T75 cell-culture flask, robot-driven acceleration and measured longitudinal wall-elevation trajectories, but the public record does not establish the particular internal wetted dimensions, measurement coordinate, and axis polarity needed for this question [1]. Those inputs therefore remain intentionally blank in the source-provenance contract.

2 From Euler to a modal initial-value problem

Use $x \in [-L/2, L/2]$ and $z \in [-h, 0]$, gravity g, density ρ and interface tension σ. In the accelerating container frame, the linear potential-flow equations and conditions are

$$\nabla^2\phi = 0, \qquad \phi_x(\pm L/2, z, t) = 0, \qquad \phi_z(x, -h, t) = 0, \tag{1}$$

$$\eta_t = \phi_z\big|_{z=0}, \qquad \phi_t + g\eta - \frac{\sigma}{\rho}\eta_{xx} = -a(t)x \quad (z = 0). \tag{2}$$

For antisymmetric longitudinal modes

$$k_n = \frac{(2n+1)\pi}{L}, \qquad \omega_n^2 = \left(gk_n + \frac{\sigma}{\rho}k_n^3\right)\tanh(k_n h), \quad n = 0, 1, \ldots. \tag{3}$$

With $\eta(x,t) = \sum_n q_n(t)\sin(k_n x)$ and $x = \sum_n b_n \sin(k_n x)$,

$$b_n = \frac{4(-1)^n}{Lk_n^2}, \qquad \boxed{\ddot{q}_n + \omega_n^2 q_n = -k_n \tanh(k_n h) b_n a(t).} \tag{4}$$

The forcing coefficient multiplying a is dimensionless; q_n, η have units m, k_n units m^{-1}, and ω_n units s^{-1}. The solver propagates the modes with an exact constant-acceleration propagator per sampling interval rather than a finite-difference Euler time step.

Known limits and falsifiers. For $kh \ll 1$, $\omega^2 \sim ghk^2$ (without capillarity). For $kh \gg 1$, $\omega^2 \sim gk$. For steady acceleration with $\sigma = 0$ and enough modes, the equilibrium height at the positive wall approaches $-aL/(2g)$. The solver is *undamped* by construction: any measured attenuation, meniscus response, weak contact-line dynamics, and three-dimensional effects must be handled as identified model inadequacies, never covertly tuned into this Euler baseline [2, 3].

3 Input provenance and gate discipline

Gate	Qualification meaning
E0	Exact wetted L, width, fill h, sensor x, fluid parameters, axis signs, orientation and initial-equilibrium evidence; **BLOCKED** until verified.
E1	Derivation limits, modal forcing, exact propagator, physical units, regression tests.
E2	Locked input NPY and v0.4.3 proxy-development split SHA-256; no TEST/OOD water-height access.
E3	Independent from-rest forecast across the 850 previously exposed development-validation trajectories (once E0 is met).
E4	One-mode-only conditional forecast from two measured initial heights; not equivalent to a multimodal initial-state reconstruction.
E5	Comparison uses strictly matched initial-condition/forcing specifications; otherwise diagnostic only.
E6	Proxy-family-disjoint DEVELOPMENT only; publisher true-family identifiers unavailable.
E7/E8	Source-native PKLSA NOT RUN / physical SST claims UNRESOLVED.
E9	Review of max $\lvert\eta\rvert/h$, with > 0.20 flagged as a large-amplitude linearity concern.

No frequency is calibrated against the observed $\sim 2.9\,\mathrm{Hz}$ oscillator. The primary evaluation uses $q_n(0) = \dot{q}_n(0) = 0$ and measured $a(t)$ alone. The secondary evaluation explicitly consumes two observed heights *at the origin only* to initialize a *single* mode. Their RMSE statistics must never be aggregated.

4 Execution, provenance and next step

Install in the existing `F005-v0.1.0` root, preserving v0.1.0–v0.4.3 outputs. Run `run_all_v050.cmd DEMO` for nonphysical synthetic verification, then populate `config/EULER_GEOMETRY_REQUIRED.json` from an actual source record and run `run_all_v050.cmd`. Without sourced geometry, expect exit code 2 and `BLOCKED_GEOMETRY_UNVERIFIED`; that is an evidence-integrity *success*, not a failed solver.

Independent geometry/measurement information may be recoverable from the linked Camilotto thesis [4], but no numerical dimensions were taken from the uninspected thesis. A manufacturer's generic T75 exterior dimensions are not an admissible substitute for the actual internal wetted domain. Only after E0 and E1–E9 are reviewed should one consider a geometry-corrected 3D or viscous Euler/Navier–Stokes benchmark, and later an independent PKLSA comparison.

References

[1] N. Giulietti, M. Bedodi, D. Ghavami, D. Camilotto, and H. Giberti (2026), "Robot-executed longitudinal sloshing dataset for rectangular tank liquid handling," Zenodo v1.0.0, https://doi.org/10.5281/zenodo.20039757.

[2] O. M. Faltinsen and A. N. Timokha (2009), *Sloshing*, Cambridge University Press, ISBN 9780521881111, https://www.cambridge.org/9780521881111.

[3] O. M. Faltinsen, O. F. Rognebakke, I. A. Lukovsky, and A. N. Timokha (2000), "Multidimensional modal analysis of nonlinear sloshing in a rectangular tank with finite water depth," *Journal of Fluid Mechanics* **407**, 201–234, https://doi.org/10.1017/S0022112099007569.

[4] D. Camilotto (2026), "A comprehensive data-driven framework for robotic liquid sloshing control: from extensive experimental model validation and enhancement to trajectory optimization," Master's thesis, Politecnico di Milano, https://hdl.handle.net/10589/253216.